

Workflow on NPU

Vector Unit

QB Decomposition

Lanczos Algorithm

Cube Unit

The Original Matrix Reduction
 $C = Q^T A, E = CC^T$

Obtain Eigen Decomposition
 $E = U_1 \Sigma_1 U_1^T$

The Resulting Matrices
 $U = QU_1, \Sigma = \Sigma_1^{1/2}$
 $V^T = \Sigma_1^{-1/2} U_1^T C$

Workflow on CPU

Symmetric QR
Iterative Algorithm
with Displacement
for Tridiagonal
Matrices

